

# A Holographic View of the QCD Axion

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## Abstract

These notes meticulously transcribe a seminar on combining warped extra dimensions with the Peccei-Quinn mechanism to solve the strong CP problem. The framework maps the axion quality problem into a holographic AdS<sub>5</sub> setting via the AdS/CFT dictionary. It explicitly explores symmetry breaking mechanisms across the bulk and branes, the 4D/5D duality of gauge axions, and investigates whether 5D small instantons can dynamically enhance the QCD axion mass.

## 1 The Strong CP Problem

The strong CP problem originates from the CP-odd topological term in the QCD Lagrangian:

$$\mathcal{L}_{\text{QCD}} \supset \theta_{\text{QCD}} \frac{\alpha_s}{8\pi} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}. \quad (1)$$

Because of the chiral anomaly, the physically observable parameter is basis independent and must include the phases of the quark mass matrix:

$$\bar{\theta} = \theta_{\text{QCD}} + \arg(\det M_q) \quad (2)$$

This effective term generates an observable effect on the neutron electric dipole moment, given by  $d_n \simeq (5 \times 10^{-16} e\text{-cm}) \bar{\theta}$ . Experimental bounds demand that  $\bar{\theta} \leq 10^{-10}$ .

Why is it so small? As Witten (1979) [Wit79] noted, this extreme fine-tuning does not appear to be "anthropic"—our universe is perfectly possible for  $\bar{\theta} \sim \mathcal{O}(1)$ .

## 2 The Peccei-Quinn Mechanism and KSVZ Model

To solve this dynamically, the Peccei-Quinn (PQ) mechanism introduces a spontaneously broken global  $U(1)_{\text{PQ}}$  symmetry [PQ77, Wei78, Wil78].

The axion field is parameterized via a complex scalar:

$$\Phi = \frac{1}{\sqrt{2}}(f_a + \rho)e^{ia/f_a} \quad (3)$$

Below the symmetry breaking scale, the radial mode is integrated out, leaving the effective Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu a)^2 + \frac{\alpha_s}{8\pi} \left( \frac{a}{f_a} + \bar{\theta} \right) G\tilde{G} + \frac{1}{4} a g_{a\gamma\gamma} F\tilde{F} + \frac{1}{f_a} J^\mu \partial_\mu a \quad (4)$$

In the KSVZ framework [Kim79, SVZ80], the scalar potential is  $V(\Phi) = -m_{\text{PQ}}^2 |\Phi|^2 + \lambda_{\text{PQ}} |\Phi|^4$ , yielding a VEV of  $\langle \Phi \rangle = f_a/\sqrt{2}$ , where  $f_a = \sqrt{m_{\text{PQ}}^2/\lambda_{\text{PQ}}}$ . Heavy quarks acquire a mass via the Yukawa coupling:

$$\Delta\mathcal{L} = y\bar{\psi}\Phi\psi + \text{h.c.} \implies m_\psi \sim y \frac{f_a}{\sqrt{2}} \quad (5)$$

Integrating out these Dirac fermions naturally generates the dimension-5 axion-gluon coupling.

### 3 The Axion Quality Problem and Guiding Questions

Gravitational violation of the global  $U(1)_{\text{PQ}}$  symmetry induces higher-dimensional, Planck-suppressed operators of the form:

$$\Delta\mathcal{L}_{\text{grav}} = \frac{|C_n|e^{i\delta_n}}{M_P^{n-4}} \Phi^n + \text{h.c.} \quad (6)$$

This modifies the axion potential to:

$$V(a) \simeq V_{\text{QCD}}(a) - \frac{|C_n|f_a^n}{M_P^{n-4}} \cos\left(\frac{na}{f_a} + \delta_n\right) \quad (7)$$

Because of the arbitrary phase  $\delta_n$ , the minimum of the potential shifts such that  $\theta_{\text{eff}} \neq 0$ . To solve the axion quality problem, these terms must be suppressed to a very high order ( $n \geq 10$ ) [Ran92, Dob96, Che01].

This leads to three fundamental questions:

1. What is the origin of the PQ breaking potential and the spontaneous breaking scale  $f_a$ ?
2. How to solve the axion quality problem? Is there a connection to GUT or SUSY?
3. Can the QCD axion mass be different from  $m_{a,\text{QCD}}$ ?

To address these questions, we use a warped dimension.

### 4 Warped Extra Dimensions (AdS<sub>5</sub>)

To generate a hierarchy of scales and small couplings, we introduce a warped 5D dimension [CGN19]. The 5D metric is defined by:

$$ds^2 = \frac{1}{(kz)^2} (\eta_{\mu\nu} dx^\mu dx^\nu + dz^2) = g_{MN} dx^M dx^N \quad (8)$$

where the IR and UV scales are related by  $\Lambda_{\text{IR}} = \Lambda_{\text{UV}} e^{-\pi k R}$ .

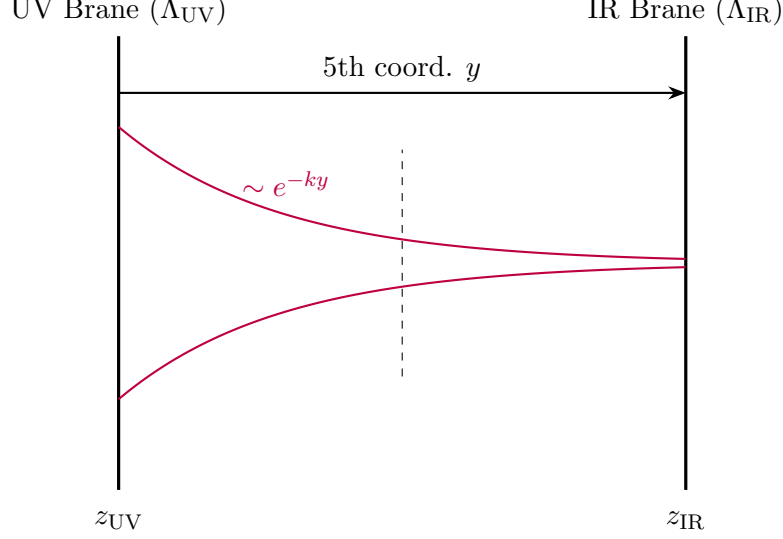


Figure 1: Warped AdS<sub>5</sub> geometry showing the UV and IR branes with a localized bulk wavefunction.

#### 4.1 Symmetry Breaking in the Bulk

The bulk scalar field  $\Phi = \frac{1}{\sqrt{2}}\eta(z)e^{ia(x,z)/f_a}$  has a bulk mass of  $m_\Phi^2 = \Delta(\Delta-4)k^2$ . Its background VEV profile is parameterized to capture two distinct symmetry-breaking mechanics:

$$\eta(z) = k^{3/2} (v(kz)^{\Delta-4} + \sigma(kz)^\Delta) \quad (9)$$

- $v$ : Governs the explicit breaking due to gravity, localized on the UV brane.
- $\sigma$ : Governs the spontaneous breaking of the axion symmetry propagating through the bulk.

#### 4.2 No UV PQ-Breaking ( $v = 0$ )

If there is no explicit UV breaking ( $v = 0$ ), the global shift symmetry  $a^{(0)}(x^\mu) \rightarrow a^{(0)}(x^\mu) + \text{const}$  is exact, resulting in a perfectly massless axion. The decay constant evaluates to:

$$f_a^{(0)}(z) = f_a^{(0)}(z_{UV}) \simeq \frac{z_{IR}}{\sigma_0} \sqrt{\Delta-1} \quad (z_{IR} \gg z_{UV}) \quad (10)$$

#### 4.3 UV PQ-Breaking ( $v \neq 0$ )

When gravity induces explicit breaking on the UV brane, the potential includes an interaction:

$$U_{UV}(\Phi) \supset -2l_{UV}k^{5/2}\eta \cos\left(\frac{a}{f_a}\right) = -2l_{UV}k^{5/2}\eta \left(1 - \frac{a^2}{2f_a^2} + \dots\right) \quad (11)$$

This explicitly breaks the symmetry, suppressing the axion decay constant such that:

$$f_a^{(0)}(z_{UV}) \simeq \frac{4z_{IR}\sqrt{\Delta-1}}{l_{UV}}(\Delta-2) \left(\frac{z_{UV}}{z_{IR}}\right)^\Delta \quad (z_{UV} \ll z_{IR}, \Delta > 4) \quad (12)$$

The resulting UV axion mass is heavily suppressed for large scaling dimension  $\Delta$ :

$$(m_a^{(UV)})^2 = \frac{6l_{UV}v(\Delta-2)}{k(\Delta-4+l_{UV})} \left( \frac{k\sqrt{\Delta-1}}{v} \right)^\Delta \left( \frac{f_a}{\Lambda_{UV}} \right)^{\Delta-4} f_a^2 \quad (13)$$

Requiring  $(m_a^{(UV)})^2 \leq 10^{-10}(m_a^{(QCD)})^2$  guarantees protection for  $\Delta \geq 10$ .

## 5 AdS/CFT Dictionary: 4D and 5D Duality

The holographic interpretation provides a powerful duality mapping between 5D AdS space and strongly coupled 4D gauge theories. The explicit AdS/CFT dictionary is:

- **5D local gauge symmetry**  $\leftrightarrow$  **4D global symmetry**
- **5D field**  $\Phi$  with  $m_\Phi^2 = \Delta(\Delta-4)k^2 \leftrightarrow$  **4D operator**  $\mathcal{O}$  with  $\dim \mathcal{O} = \Delta$
- **5D scale**  $\Lambda_{UV}e^{-\pi kR} = \Lambda_{IR} \leftrightarrow$  **4D scale**  $\Lambda_{IR} = \Lambda_{UV}e^{-\frac{8\pi^2}{b_{\text{CFT}}g^2}}$

Under this holographic interpretation, the 5D Axion corresponds exactly to a 4D composite axion featuring an accidental global PQ symmetry [Kim85, CK85]. The elementary PQ scalar becomes a dynamical fermion condensate  $\langle \bar{\psi}\psi \dots \psi \rangle$ . Consequently, the axion quality problem maps directly between dimensions:

$$\frac{c_n}{M_P^{n-4}} \Phi^n \leftrightarrow \frac{c_\Delta}{M_P^{\Delta-4}} \mathcal{O} \sim \frac{c_\Delta}{M_P^{\Delta-4}} (\psi_i \psi_j \dots \psi_k) \quad (\Delta \geq 10) \quad (14)$$

These models better incorporate SM features by introducing naturally massless exotic quarks, making the SM part of the strong dynamics, and providing a connection to GUT/supersymmetry breaking [GMQ25, ST25, GMNQ25].

## 6 5D Gauge Axion in Warped GUT

In a warped Grand Unified Theory (GUT) [CG25], the SM fermions  $\{10_i, \bar{5}_i\}$  and Higgs  $\{24_\Phi, 5_H, \bar{5}_H\}$  are embedded in  $SU(5)_{\text{GUT}} \times U(1)_{\text{PQ}}$ . The fields are represented by  $A_M^{(\text{GUT})}$  and  $A_M^{(\text{PQ})}$ .

The QCD axion resides in the 5th component, defined by  $\theta(x) = \frac{a(x)}{f_a} = \oint dy A_5(x, y)$ . This guarantees an exact shift symmetry under  $U(1)_c^{(1)}$ :

$$A^{(\text{PQ})} \rightarrow A^{(\text{PQ})} + \Lambda_c^{(1)}, \quad d\Lambda_c^{(1)} = 0 \implies A_5^{(\text{PQ})} \rightarrow A_5^{(\text{PQ})} + \lambda \quad (15)$$

The axion quality violation due to brane-localized charged matter takes the suppressed form:

$$\Delta V(\theta) \sim M_5^4 e^{-2\pi k R_1} e^{-\pi R \sqrt{M_5^2 + 4k^2}} \cos\left(\frac{q\theta}{2} + \delta\right) \quad (16)$$

## 6.1 Gauge Coupling Unification

In warped  $\text{AdS}_5$ , the gauge coupling unification is preserved with a logarithmic scaling:

$$\frac{2\pi}{\alpha_i(\mu)} = \frac{2\pi}{\alpha_i(m_{\text{KK}})} - (b_i^{(\text{SM})} - b_i^{(\text{CFT})}) \log\left(\frac{m_{\text{KK}}}{\mu}\right) \quad (17)$$

Using MSSM beta functions  $(3, -1, -33/5)$ , the warped 5D gauge axion is highly compatible with grand unification scales  $f_a \geq 10^9$  GeV [Ben25].

## 6.2 Possible Dual 4D Description

This setup boasts a precise 4D dual mapping:

- 5D bulk  $U(1) \leftrightarrow$  4D global  $U(1)_B$
- IR breaking of  $U(1)$  ( $\int_{\text{UV}}^{\text{IR}} dy A_5$ )  $\leftrightarrow$  Spontaneously broken  $U(1)_B$  ( $\langle B \rangle \neq 0, \arg(\mathcal{B})$ )
- 5D locality (no charged matter)  $\leftrightarrow \dim B \sim N \rightarrow \infty$

The PQ symmetry is spontaneously broken by an infinite dimension operator. A wrapped D3 particle is charged under  $U(1) = \text{baryon}$ . For 4D SQCD where  $N_f = N_c = N$ , we have  $\det M - B\bar{B} = \Lambda^{2N}$ , with  $U(1)_{\text{PQ}} = U(1)_B$  and  $B \sim Q^N$ .

## 7 Enhancing the Axion Mass and Small Instantons

The QCD axion mass is derived from the topological susceptibility  $\mathcal{T}$ :

$$m_a^2 f_a^2 = \mathcal{T} \equiv -i \int d^4x \langle 0 | T \left[ \frac{\alpha_s}{8\pi} G\tilde{G}(x), \frac{\alpha_s}{8\pi} G\tilde{G}(0) \right] | 0 \rangle \quad (18)$$

In the dilute instanton gas approximation:

$$\mathcal{T} \propto \int \frac{d\rho}{\rho^5} C[3] \left( \frac{2\pi}{\alpha_s(1/\rho)} \right)^6 e^{-\frac{2\pi}{\alpha_s(1/\rho)}} \quad (19)$$

Because QCD is asymptotically free, this is suppressed by fermion zero modes  $(\rho m_f)^{N_f}$ , yielding the standard mass

$$m_{a,\text{QCD}}^2 = \frac{m_u m_d}{(m_u + m_d)^2} \frac{m_\pi^2 f_\pi^2}{f_a^2}. \quad (20)$$

To enhance this mass, we can change the QCD coupling in the UV (via small instantons) or close fermion loops with Higgs configurations ( $\kappa_f$ ).

### 7.1 5D Small Instantons in Flat Space

For flat 5D space  $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu + dy^2$ , the localized instanton profile is

$$A_\mu^{(I)a}(x) = \frac{2\eta_{a\mu\nu}(x - x_0)_\nu}{(x - x_0)^2 + \rho^2} \quad (21)$$

$A_5^a = 0$  [PS02]. This gives a finite action  $S_5^{(I)} = \frac{8\pi^2 R}{g_5^2} = \frac{2\pi}{\alpha_s}$ . Including Kaluza-Klein fluctuations [DDG98]:

$$S_{\text{eff}} = \frac{2\pi}{\alpha_s(1/R)} - 3\xi(R/\rho)\frac{R}{\rho} + b_0 \ln \frac{R}{\rho} \quad (22)$$

The power-law term ( $-\frac{R}{\rho}$ ) overcomes the suppression, allowing small 5D instantons to dominate for  $1/R \gtrsim 100$  TeV [GKPS20]. This holds up to  $\Lambda_5 R \leq \frac{6\pi}{\alpha_s}$ .

## 7.2 Holographic Obstruction in AdS<sub>5</sub>

When embedded in warped AdS<sub>5</sub> [GP21], the coupling behaves as:

$$\frac{2\pi}{\alpha_s(1/\rho)} = \frac{2\pi}{\alpha_s(M_{\text{IR}})} - \left(b_0 + \frac{8\pi^2}{g_5^2 k}\right) \log \left(\frac{\rho^{-1}}{M_{\text{IR}}}\right) \quad (23)$$

Applying the classical bulk profile

$$A_\mu^{(cl)}(x, z) = -i(\partial_\mu \Omega) \Omega^{-1} \frac{x^2}{x^2 + \rho^2}, \quad (24)$$

the action resolves to

$$S_5^{(I)} = \frac{8\pi^2}{g_5^2 k} \log \frac{M_{\text{UV}}}{M_{\text{IR}}}. \quad (25)$$

Holographically, the CFT constituents must be bifundamental colored matter. Two possibilities exist:

1. **Colored fermion constituents:** This results directly in chiral suppression from CFT fermions  $(\rho M_{\text{IR}})^{b_{\text{CFT}}}$ , meaning no axion mass enhancement.
2. **Colored scalar constituents:** This bypasses chiral suppression and enhances the mass. However, topological charge conservation dictates that no localized instanton exists for this boundary condition.

## 7.3 Holographic Obstruction in AdS<sub>5</sub>

When embedded in warped AdS<sub>5</sub>, we must evaluate the QCD instantons in the 5D bulk [GP21]. The coupling behaves as:

$$\frac{2\pi}{\alpha_s(1/\rho)} = \frac{2\pi}{\alpha_s(M_{\text{IR}})} - \left(b_0 + \frac{8\pi^2}{g_5^2 k}\right) \log \left(\frac{\rho^{-1}}{M_{\text{IR}}}\right) \quad (26)$$

Applying the classical 5D test profile  $A_\mu^{(cl)}(x, z) = -i(\partial_\mu \Omega) \Omega^{-1} \frac{x^2}{x^2 + \rho^2}$  (where  $\Omega \in SU(2)$ ) and  $A_5(x, z) = 0$ , we integrate the classical field strength over the warped bulk to find the finite action:

$$\begin{aligned} S_5^{(I)}[A_\mu^{(cl)}(x, z)] &= -\frac{1}{2g_5^2} \int d^4x \int_{1/M_{\text{UV}}}^{1/M_{\text{IR}}} dz \sqrt{g} \text{Tr} \left[ G_{MN}^{(cl)} G^{MN(cl)} \right] \\ &= \frac{8\pi^2}{g_5^2 k} \log \frac{M_{\text{UV}}}{M_{\text{IR}}} = \frac{2\pi}{\alpha_s(M_{\text{IR}})} \neq \frac{2\pi}{\alpha_s(1/\rho)} \end{aligned} \quad (27)$$

Because the result is fixed at  $\frac{2\pi}{\alpha_s(M_{\text{IR}})}$  and does not equal  $\frac{2\pi}{\alpha_s(1/\rho)}$ , the instanton action fails to scale with the instanton size  $\rho$ .

Holographically, this indicates a fundamental obstruction. The CFT constituents must be bifundamental colored matter. There are two possibilities:

1. **Colored fermion constituents:** This results directly in chiral suppression from CFT fermions  $(\rho M_{\text{IR}})^{b_{\text{CFT}}}$ , meaning no axion mass enhancement.
2. **Colored scalar constituents:** This bypasses chiral suppression and enhances the mass. However, strict topological charge conservation dictates that no localized instanton exists for this boundary condition.

## 8 Summary

- The axion quality problem can be fully solved in a 5D warped dimension, perfectly dual to a 4D composite axion with an accidental PQ symmetry.
- This warped framework elegantly explains fermion mass hierarchies and naturally incorporates GUT and axion isocurvature scales.
- Flat 5D small instantons can enhance the axion mass while protecting the strong CP solution. However, a warped AdS<sub>5</sub> instantiation faces an inescapable obstruction dictated by the nature of colored constituents in the dual CFT.

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